LIFE & ANNUITIES · RETIREMENT CONSULTING · ESTATE STRATEGY

Planning Platform

Methodology and Verification

A reference for independent review. It states what the platform computes, where every figure comes from, what it deliberately does not model, and how each claim here can be checked.

Edwards Financial & Associates
Joshua Edwards, Consultant
CA Insurance License #0G52544 · NPN #13411537
edwardsfinancialassociates.com

Prepared October 2026
Parameter set effective for the 2026 tax year

1. What this is

An interactive retirement, tax and estate planning platform. It projects one household year by year from today to the end of its planning horizon, and presents that single projection through thirty-three views across eight modules: Dashboard, Investment, Retirement, Insurance, Education, Tax, Estate and Profile.

It was built in-house. It is not a licensed or white-labelled copy of any commercial platform, and no third-party planning software contributed code, content or parameter values to it.

> **The governing principle.** Every statutory figure traces to the document that sets it, and that document is named on the output. Where a figure is a house view rather than a rule, the platform says so rather than presenting it as a fact. Where a figure cannot be sourced, the platform refuses to produce a client report rather than filling the gap.

2. How the engine works

One ledger produces every number. For each year of the plan, in order:

- Age each person and test retirement, death and horizon triggers.
- Collect inflows: earned income, Social Security, pension, annuity and other income. A Social Security benefit carrying a stated figure at full retirement age is recomputed from that figure and the claiming age, and the earnings test is applied where it bites.
- Inflate every stream by its own index. Five indices are carried separately: general, health care, education, Social Security and tax. Health and education do not track general inflation.
- Collect outflows: living expenses, housing, health care, insurance premiums, goals and debt.
- Compute required minimum distributions, then discretionary withdrawals in the chosen sequence.
- Compute tax: federal ordinary income with capital gains stacked on top, the two-tier taxation of Social Security benefits, the net investment income tax, payroll tax, and state tax.
- Apply savings and employer contributions to the correct account types.
- Grow each account by its asset-class-weighted return, splitting that return into interest, dividend, realised capital gain and unrealised appreciation so each piece is taxed correctly.
- Carry balances forward and record every line.

The simulation layer runs that loop repeatedly with returns drawn at random from each asset class, correlated across classes using a Cholesky decomposition of the published correlation matrix. A fresh correlated draw is taken for *every year of every run*, not once per run. Probability of success is the share of runs that reach the end of the horizon without running out.

Why the return is split four ways

A single total return number cannot be taxed correctly. Interest is ordinary income, qualified dividends and realised gains are taxed on their own schedule, and unrealised appreciation is not taxed at all until it is sold. A plan built on one blended number will misstate after-tax outcomes for the whole horizon, and the error compounds. Each asset class therefore carries all four components, and each component is sourced separately.

3. Where every figure comes from

Twenty-nine parameter blocks across nine files, each carrying the document it was read from:

PARAMETER SET	BLOCK	SOURCE DOCUMENT	LOCATION
Capital market assumptions		Horizon Actuarial Services, Survey of Capital Market Assumptions, 2026 Edition	https://www.horizonactuarial.com/survey-of-capital-market-assumptions
Long-term care and retiree health costs	longTermCare	Cost of Care Survey 2025	https://assets.carescout.com/x/8fcb50422f/282102.pdf
	retireeHealthEstimates.fidelity	Fidelity Investments Shares 25th Annual Retiree Health Care Cost Estimate	https://alumni.fidelity.com/news/11553433
	retireeHealthEstimates.ebri	Projected Savings Medicare Beneficiaries Need for Health Expenses in Retirement up Again in 2025	https://www.ebri.org/health/publications/fast-facts/content/full/projected-savings-medicare-beneficiaries-need-for-health-expenses-in-retirement-up-again-in-2025
Federal income and estate tax			
	estate	Instructions for Form 706, revised July 2026, Table A Unified Rate Schedule	https://www.irs.gov/instructions/i706
	contributionLimits.announced2027	Rev. Proc. 2026-24, 2027 HSA and HDHP amounts	https://www.irs.gov/pub/irs-drop/rp-26-24.pdf
Medicare premiums and IRMAA			
	partD.announced2027	Medicare Part D 2027 National Average Monthly Bid Amount Information	https://cms.gov/newsroom/fact-sheets/medicare-part-d-2027-national-average-monthly-bid-amount-information
Public pension systems	CalSTRS		
	CalPERS		
Required minimum distributions		TD 9930, Updated Life Expectancy and Distribution Period Tables Used for Purposes of Determining Minimum Required Distributions, 85 FR 72472, November 12, 2020	https://www.govinfo.gov/content/pkg/FR-2020-11-12/pdf/2020-24723.pdf
Social Security		Normal Retirement Age, SSA Office of the Chief Actuary, with the per-rule sources recorded on each block below	https://www.ssa.gov/oact/progdata/nra.html
	earlyReduction	Benefit Reduction for Early Retirement, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/quickcalc/earlyretire.html
	pia	Primary Insurance Amount, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/cola/piaformula.html
	cola	Social Security Announces 3.6 Percent Benefit Increase for 2027	https://www.ssa.gov/news/en/press/releases/2026-10-14.html
	earningsTest	Exempt Amounts Under the Earnings Test, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/cola/rtea.html

PARAMETER SET	BLOCK	SOURCE DOCUMENT	LOCATION
	spousal	Benefits for Spouses, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/quickcalc/spouse.html
	survivor	What you could get from Survivor benefits, SSA	https://www.ssa.gov/survivor/amount
	survivor.waypoints	What you could get from Survivor benefits, SSA	https://www.ssa.gov/survivor/amount
	contributionAndBenefitBase	Contribution and Benefit Base, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/cola/cbb.html
	maximumBenefitAtFullRetirementAge	Workers with Maximum-Taxable Earnings, SSA Office of the Chief Actuary	https://www.ssa.gov/oact/cola/examplemax.html
	windfallEliminationProvision	Public Law 118-273, U.S. Government Publishing Office	https://www.govinfo.gov/content/pkg/PLAW-118publ273/html/PLAW-118publ273.htm
State income tax	CA	Tax News, October 2026, section 2026 Indexing, Franchise Tax Board	https://www.ftb.ca.gov/about-ftb/newsroom/tax-news/index.html
	CA.earlyDistributionAdditionalTax	2025 Instructions for Form FTB 3805P, Additional Taxes on Qualified Plans	https://www.ftb.ca.gov/forms/2025/2025-3805p-instructions.html
	TX		
	FL		
	NV		
	WA		

4. Verified figures against house views

The platform distinguishes the two and never blurs them.

Verified against a primary source

Federal tax brackets, standard deduction, capital gains thresholds, the net investment income tax, contribution limits and the estate exclusion. Required minimum distribution ages and the Uniform Lifetime Table. Social Security full retirement age, early reduction and delayed credit factors, primary insurance amount bend points, the cost-of-living adjustment, the earnings test, and spousal and survivor rules. Medicare premiums, deductibles, income-related adjustment tiers and late enrolment penalties. California income tax, including the additional tax on early distributions. CalSTRS and CalPERS age factors and benefit formulas.

House views, disclosed as such

Capital market assumptions, inflation indices, portfolio turnover, and the share of dividends that are qualified. No authority publishes a correct expected return. The platform therefore names the source it uses and prints that source on the face of the assumption table, so a reader can see whose view of the future is being shown.

The current set is the Horizon Actuarial Services Survey of Capital Market Assumptions, 2026 Edition, published August 2026: the average across forty-three investment advisors, over a ten-year horizon, stated as geometric nominal returns in US dollars, with the full seventeen-by-seventeen correlation matrix. It was selected because it is the only widely published set that provides

expected returns, volatilities and correlations together. The dividend and interest components of that return are sourced separately from the index providers and the US Treasury.

5. What the platform refuses to do

Three mechanical gates run before anything reaches a client.

A parameter set with no sourced file behind it	**Blocks the report**
A statutory block carrying no citation	**Reported by name**
A correlation matrix that is not positive semi-definite	**Refuses to simulate**
A projection that goes non-finite	**Reported as a failure**
A return component below zero	**Blocks the report**
An asset class with no volatility	**Blocks the report**
An asset class with an unsourced income split	**Flagged by name**

The third gate deserves a note. A correlation matrix can be symmetric, carry 1.00 down its diagonal, and hold every entry between minus one and one, and still describe a set of relationships no market can have. Such a matrix cannot be decomposed. Left unchecked it produces values that are not numbers, and because every comparison against a value that is not a number is false, a shortfall test never fires and the output reports complete success. The platform tests for this explicitly and refuses to run, naming the asset class at which the decomposition failed.

6. Test coverage

Three hundred and fourteen automated tests across four suites, all passing.

Engine behaviour, tax arithmetic, triggers, scenarios	**117 tests**
Published figures reproduced exactly	**140 tests**
Capital market coherence and the client gate	**38 tests**
Portfolio arithmetic against closed-form answers	**19 tests**

The second suite is the one that matters most to a reviewer. It does not test that the code runs; it tests that the code reproduces figures published by the issuing authority. It checks the bracket base amounts for all four filing statuses against the revenue procedure, the Social Security reduction and delayed credit percentages against the Office of the Chief Actuary's own stated examples, and the CalSTRS age factor against CalSTRS's own worked examples, including the career factor cap.

The fourth suite pins portfolio arithmetic to answers with a closed form, including the case where two perfectly opposed assets cancel to zero volatility at a specific weighting. Portfolio volatility is computed as the square root of the weighted sum of every pairwise covariance, not as a weighted average of the parts. The difference on a sixty-forty mix is approximately 1.6 percentage points of volatility, and treating it as an average would discard the entire measurable benefit of diversifying.

7. What is deliberately not modelled

Stated plainly so that a reviewer does not have to infer it.

- **Alternative minimum tax.** Not computed. Households likely to be exposed to it should be reviewed outside the platform.
- **State tax beyond the loaded states.** California is modelled in full. Texas, Florida, Nevada and Washington are modelled as having no state income tax on this income. Any other state returns a reported gap rather than a zero.
- **Local and municipal income tax.** Not computed.
- **Individual security holdings.** Accounts carry balances and asset-class weights. Sector, style and concentration analysis require holdings and report that they are unavailable without them.
- **Real-time account aggregation.** There is none. Figures are entered or imported.
- **Insurance product mechanics.** Carrier charges, caps, participation rates, bonus vesting and index crediting are not simulated. Any illustration figure must come from the carrier's own illustration system.
- **REIT distribution character.** A real estate distribution splits into ordinary income, qualified business income and return of capital. That split is not yet verified, and the whole yield is currently modelled as a dividend. This is an approximation and is noted in the parameter file.
- **Six asset classes carry no sourced income split.** US small and mid cap, hedge funds, commodities, infrastructure, private equity and private debt. Their whole return currently sits in appreciation, which understates the tax they generate in a taxable account. Each is flagged by name in the platform rather than passing silently.

8. Where the data lives

A plan edited in the platform is held in the browser on the device being used. It is not transmitted, not stored on a server, and not visible to anyone else. It can be exported to a file and imported on another device. Because browser storage is cleared by ordinary actions such as clearing site data, it is not a system of record, and the platform says so on every editing page.

9. Open items

- The capital market source could not be parsed from its original file in the build environment. Its tables were transcribed by two independent readings that agreed on every one of the eighteen return rows and all one hundred and fifty-three matrix cells, and a third-party summary matched on spot checks. The published exhibit should be confirmed directly before any client-facing use.
- The threshold at which the Medicare income-related adjustment begins for a married person filing separately is stated differently in two official publications. Both figures are recorded and the conflict is flagged rather than resolved by preference.

- The required minimum distribution age for a person born in 1959 rests on a proposed regulation, because the statute as written yields two answers at once and the defining paragraph is reserved. The platform records this rather than choosing silently.
- CalSTRS does not publish its option factors. The platform declines to compute a Modified Benefit rather than estimating one.

10. How to verify this yourself

- Every parameter file carries the source document's title, its web address and its effective year on each block. Open the file, open the source, compare the figure.
- The assumption table inside the platform prints its source on the page, not in a footnote.
- The test suites can be run and read. Each test states in plain language what it asserts and why, and the suite that reproduces published figures names the publication for each one.
- Any projection can be exported year by year, so an independent model can be run against the same inputs and the two compared line by line.

> **A note on scope.** This platform is a modelling and illustration tool. It does not constitute tax, legal or investment advice, and its projections are not guarantees. Tax treatment described here reflects current law and is subject to change. Any figure intended for a client decision should be confirmed against the primary source and, where it concerns tax or legal treatment, reviewed by a qualified professional in that field.